

- 5 Radio waves can be used to measure the distance between Earth and the planet Jupiter.

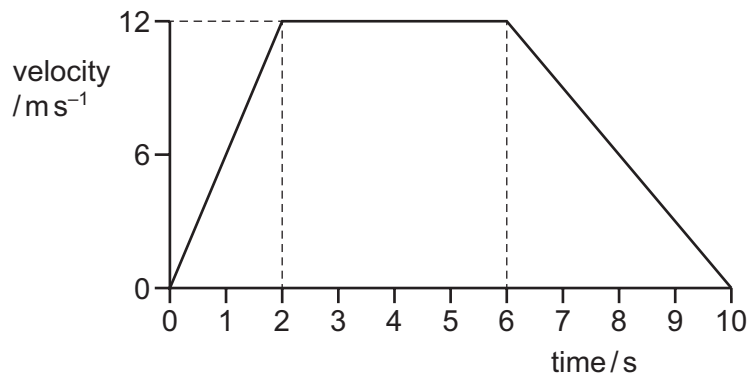
A pulse of radio waves is emitted from the surface of Earth. The pulse reflects from the surface of Jupiter and is detected again on Earth.

The time between emitting and receiving the pulse is 3960 s.

What is the distance between Earth and Jupiter?

- A $5.94 \times 10^8 \text{ km}$
- B $1.19 \times 10^9 \text{ km}$
- C $5.94 \times 10^{11} \text{ km}$
- D $1.19 \times 10^{12} \text{ km}$

- 6 The graph shows the variation with time of the velocity of a car.



Which statement is correct?

- A The car accelerates for 2 s, then stops for 4 s and then reverses.
 - B The car accelerates at 12 m s^{-2} for 2 s.
 - C The car travels a distance of 36 m in the first 4 s.
 - D The car travels a distance of 48 m in the last 4 s.
- 7 A solid object of mass 1.0 kg falls vertically downwards in a vacuum.

When the speed of the object is 60 m s^{-1} , an additional constant force of 50 N suddenly starts to act vertically upwards on the object.

What is the speed of the object 2.0 s after the additional force starts to act?

- A 20 m s^{-1}
- B 40 m s^{-1}
- C 80 m s^{-1}
- D 100 m s^{-1}